99.500 Applied Mathematics for Engineering

Spring Semester AY2026

Ordinary Differential Equation

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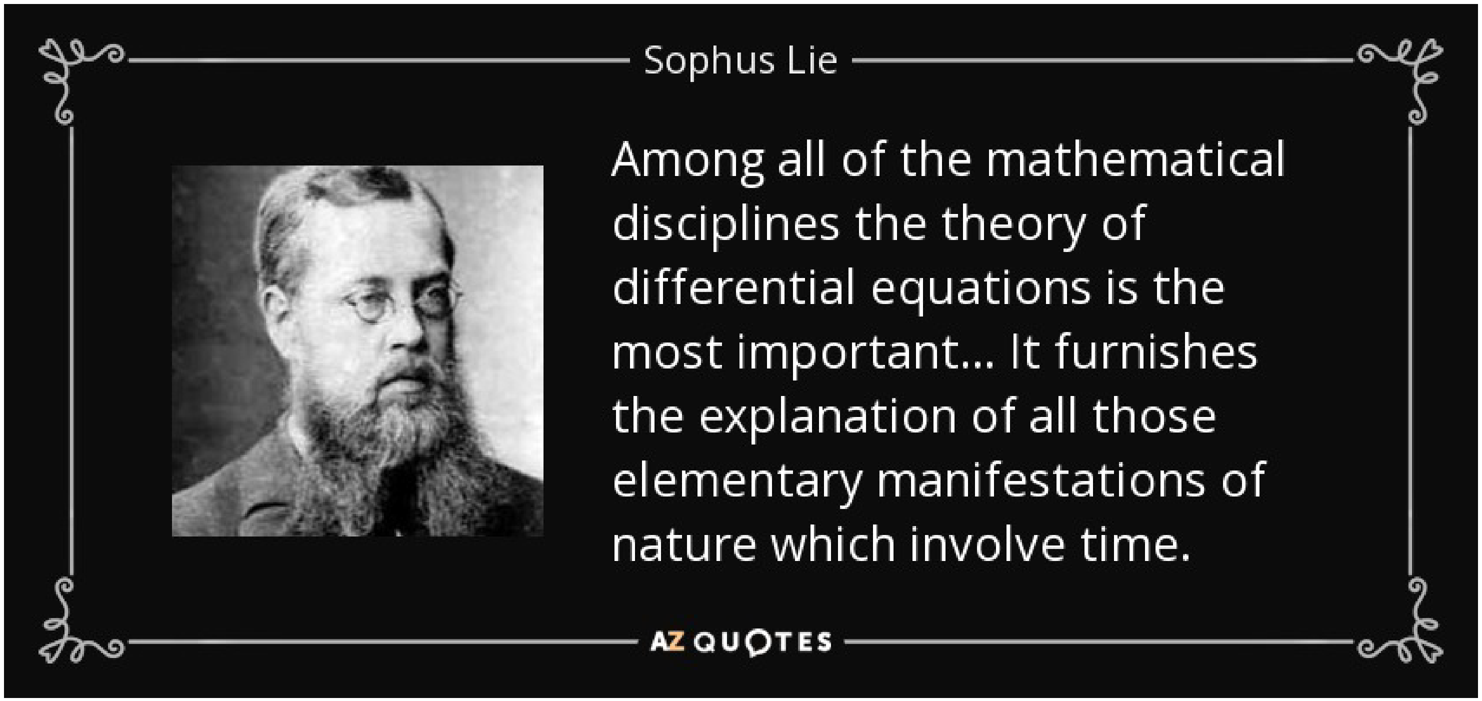
TECHNOLOGY AND DESIGN


Sophus Lie
Among all of the mathematical disciplines the theory of differential equations is the most important... It furnishes the explanation of all those elementary manifestations of nature which involve time.
AZ QUOTES

Week 3 Outline

- 1st-order linear ODE
- 2nd-order linear ODE
- Homogeneous vs non-homogeneous ODE
- Particular solution
- Mid-term project instruction

Definitions

$\frac{dy}{dx}$ $\frac{d^2y}{dx^2}$ Indep.

ODE: a relation containing one real variable x, the real dependent variable y, and some of its derivatives y', y'', $\cdots$, $y^{(n)}$, $\cdots$, with respect to x. The order of an ODE: the order of the highest derivative that occurs in the equation. Thus, an n-th order ODE has the general form

order n.

$$F(x, y, y', y'', \ldots, \boxed{y^{(n)}}) = 0$$

$y \cdot y'$ X $y' \cdot y''$ X $(y'')^2$ X

Linear ODE: the dependent variable y and all its derivatives appear to the power of one (are not multiplied or divided by each other) and are not composed with any non-linear functions of y or its derivatives. The general form of an n n-th order linear ODE is:

$\sin(y)$ X $e^{y'}$ X

derivative

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

$(x^2+1)\frac{d^2y}{dx^2}$

If $g(x) \equiv 0$, it is a **homogeneous** equation.

If $g(x) \not\equiv 0$, it is a **non-homogeneous** equation.

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Condition for ODE solution

The exact solution of n-th order ODE requires n "constraints":

1. Condition 1: If the independent variable is time, then the "constraints" are the initial conditions (ICs).
2. Condition 2: the independent variable is the position or location related, then the "constraints" are the boundary conditions (BCs).

1st-order linear ODE

to one side

rest other side

General form of a 1st-order linear ODE: $a_1(x)\underline{y'} + a_0(x)y = g(x)$

$\frac{dy}{dx}$

$$\underline{a_1(x)y' + a_0(x)y = \boxed{0}} \implies y' = -\frac{a_0(x)}{a_1(x)}y$$

$g(x)=0$

Homogeneous case

$$\implies \int \frac{dy}{y} = \int -\frac{a_0(x)}{a_1(x)}dx$$

$$\implies \ln|y| = \int -\frac{a_0(x)}{a_1(x)}dx$$

$$\implies y = Ce^{\int -\frac{a_0(x)}{a_1(x)}dx}$$

Non-homogeneous ODE: Integrating Factor

$$a_1(x)y' + a_0(x)y = g(x) \implies y' + P(x)y = Q(x)$$

$P(x)$ $= Q(x)$ $\frac{a_0(x)}{a_1(x)}$ $\frac{g(x)}{a_1(x)}$

$uy' + uPy = uQ(x)$

Integrating factor : define $\mu(x) = e^{\int P(x)dx}$

$\longrightarrow u' = \frac{du}{dx} = e^{\int p(x)dx} \cdot \frac{d}{dx}\int p\,dx = u \cdot p(x)$

$= uy' + yu' = uy' + y\,u \cdot p(x)$

$$\Rightarrow \frac{d}{dx}[\mu(x)y] = \mu(x)y' + \mu(x)P(x)y \qquad \mu(x)y' + \mu(x)P(x)y = \mu(x)Q(x)$$

$$\implies \int \frac{d}{dx}[\mu(x)y] = \int \mu(x)Q(x)$$

$$\int \frac{d}{dx}[\mu(x)y]dx = \int \mu(x)Q(x)dx \qquad \implies \mu(x)y = \int \mu(x)Q(x)dx + C$$

$$\implies y = \frac{1}{\mu(x)}\left(\int \mu(x)Q(x)dx + C\right)$$

with $e^{\int p(x)dx}$

The solution to the 1st-order linear ODE $y' + P(x)y = Q(x)$ is:

$$y = \frac{1}{e^{\int P(x)dx}}\left(\int e^{\int P(x)dx}Q(x)dx + C\right)$$

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Example

$$\int \cancel{\frac{d}{dx}}(e^{2x}y) = \int e^{3x}\,dx$$

$$e^{2x}y = \frac{1}{3}e^{3x} + C$$

P Q

$$y' + \underline{2}y = \underline{e^x}$$

Identify $P(x)$ & $Q(x)$	:	$P(x) = 2$ & $Q(x) = e^x$
Find the integrating factor $\mu(x)$	:	$\mu(x) = e^{\int 2 dx} = e^{2x}$
Multiply the ODE by $\mu(x)$	:	$\boxed{e^{2x}y' + 2e^{2x}y} = e^{2x}e^x$
Recognize the left-hand side as a derivative	:	$\frac{d}{dx}(e^{2x}y) = e^{3x}$
Integrate both sides	:	$e^{2x}y = \int e^{3x}dx + C = \frac{1}{3}e^{3x} + C$
Solve for y	:	$y = \frac{1}{3}e^x + Ce^{-2x}$

divide by e^{2x}

Apply Formula: $y = \frac{1}{e^{2x}}\left(\int e^{2x}\cdot e^{x}\,dx + C\right)$

$$\frac{1}{e^{2x}}\left(\frac{1}{3}e^{3x} + C\right)$$

2nd-order ODE: Method of Reduction to 1st Order

General form of a 2nd-order ODE : $\boxed{y''} = f(x, y, y')$

no y inside f $\longrightarrow$ $y'' = f(x, y')$ $\rightsquigarrow$ $p(x) y' + Q(x)$

$\boxed{y \text{ does not appear explicitly}}$: $y'' + P(x)y' = Q(x)$

let $\nu = y'$ & $v' = y''$ $\implies$ $\boxed{v' + P(x)v = Q(x)}$ 1st order ODE.

by slide 7.

Method of integrating factor $\implies$ $v = \dfrac{1}{e^{\int P(x)dx}} \left(\int e^{\int P(x)dx} Q(x)dx + C \right)$

$\frac{d}{dx} v = \frac{d}{dx} y'$

$\Downarrow$

$v' = y''$

$\implies$ $y' = \dfrac{1}{e^{\int P(x)dx}} \left(\int e^{\int P(x)dx} Q(x)dx + C \right)$

$\implies$ $y = \displaystyle\int \dfrac{1}{e^{\int P(x)dx}} \left(\int e^{\int P(x)dx} Q(x)dx + C \right) dx$

Example

$u(x) = ?$

$$v = \frac{1}{u(x)}\left(\int u(x)Q(x)\,dx + c\right)$$

$v' \qquad v$

$\uparrow \qquad \uparrow$

$$y'' - 3y' = x$$

$$v = y' \;\&\; v' = y'' \qquad \Longrightarrow \qquad \boxed{v' - 3v = x}$$

Method of integrating factor $\qquad \Longrightarrow \qquad v = -\frac{1}{3}x - \frac{1}{9} + Ce^{3x}$ ✓

$\frac{dy}{dx}$ "

$$\Longrightarrow \qquad y = \int -\frac{1}{3}x - \frac{1}{9} + Ce^{3x}\,dx$$

$$y = -\frac{1}{6}x^2 - \frac{1}{9}x + Ce^{3x} + D$$

2nd-order ODE: Method of Reduction to 1st Order

$y'' = f(\cancel{x}, y, y')$

no x inside

$= v \cdot \frac{dv}{dy}.$

$\boxed{x \text{ does not appear explicitly}}$: $y'' = f(y, y')$

chain rule concept.

apply $\frac{d}{dx}$

$= y' = v$

let $v = y' \implies y'' = \frac{dv}{dx} = \frac{dv}{dy}\frac{dy}{dx} = v\frac{dv}{dy}$

with v & y.

$\implies v\frac{dv}{dy} = f(y, v)$ 1st-order ODE

Find $v \implies y = \int v dx$

Example

$\Rightarrow y'' = -y = f(y, y')$

$\lambda^2 + 1 = 0$

$\lambda = \pm i$

$y = C\cancel{e^{0x}}(\cos x) + D\cancel{e^{0x}}\sin(x)$

$$y'' + y = 0$$

let $v = y' \implies v\frac{dv}{dy} = -y$

+ve -ve

$y_1 \quad y_2$ combine $z = y_1 + y_2$

$$\implies \int v\,dv = -\int y\,dy$$

$\frac{v^2}{2} = -\frac{y^2}{2} + D$

$$\implies v^2 = -y^2 + C$$

$y_1'' + y_1 = 0$ ✓

$y_2'' + y_2 = 0$ ✓

$$\implies \frac{dy}{dx} = \boxed{v} = \pm\sqrt{C - y^2}$$

$\int \frac{1}{\sqrt{C-y^2}}\,dy = \int \pm\,dx$

$$\implies \frac{dy}{dx} = \pm\sqrt{C - y^2}$$

$\sin^{-1}\frac{y}{\sqrt{C}} = \left.\begin{matrix} A - x \\ B + x \end{matrix}\right\}$

$z'' + z = (y_1 + y_2)'' + y_1 + y_2$

$$\implies y = \sqrt{C}\sin(A - x) + \sqrt{C}\sin(B + x)$$

$z'' = \underbrace{y_1'' + y_1}_{0} + \underbrace{y_2'' + y_2}_{0}$

$$\implies y = c_1 \sin(x) + c_2 \cos(x)$$

$\frac{y}{\sqrt{C}} = \sin(A - x)$

$\sin(B + x)$

$\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A.$

$y = \sqrt{C}\sin(A - x) + \sqrt{2}\sin(B + x)$

2nd-order Linear Homogeneous ODE with Constant Coefficients

$\lambda^2 \cancel{e^{\lambda x}} + a\lambda \cancel{e^{\lambda x}} + b \cancel{e^{\lambda x}} = 0$

$$y'' + ay' + by = 0$$

$(e^{\lambda x})'' = \lambda^2 e^{\lambda x}$

inside

Look for solution of the form $y = e^{\lambda x}$

$(e^{\lambda x})' = \lambda e^{\lambda x}$

$e^{\lambda x}$ is a solution $\Longleftrightarrow$ $\boxed{\lambda^2 + a\lambda + b = 0}$ → characteristic eq

solve

$$\lambda_1 = \frac{1}{2}(-a + \sqrt{a^2 - 4b})$$

$$\lambda_2 = \frac{1}{2}(-a - \sqrt{a^2 - 4b})$$

Case 1 $a^2 - 4b > 0 \Longrightarrow y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$

Case 2 $a^2 - 4b = 0 \Longrightarrow \lambda_1 = \lambda_2, y = c_1 e^{\lambda x} + c_2 x e^{\lambda x}$

$$\begin{aligned}
a^2 - 4b < 0 \quad &\Longrightarrow \quad \lambda_1 = \alpha + i\beta, \lambda_2 = \alpha - i\beta \\
&\Longrightarrow \quad y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x} \\
&\Longrightarrow \quad y = e^{\alpha x}(c_1 e^{i\beta x} + c_2 e^{-i\beta x}) \\
&\Longrightarrow \quad y = e^{\alpha x}(c_1 \cos(\beta x) + ic_1 \sin(\beta x) + c_2 \cos(\beta x) - ic_2 \sin(\beta x)) \\
&\Longrightarrow \quad y = Ce^{\alpha x} \cos(\beta x) + De^{\alpha x} \sin(\beta x)
\end{aligned}$$

$e^{i\theta} = \cos\theta + i\sin\theta$

Restricted

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Examples

$y'' + y' - 2y = 0, \underbrace{y(0) = 4, y'(0) = -5}_{BCs}$ Solution: $\lambda_1 = 1, \lambda_2 = -2$

$\lambda^2 + \lambda - 2 = 0$

$\lambda_1 = 1, \lambda_2 = -2$

$y = c_1 e^x + c_2 e^{-2x}$

$y(0) = c_1 + c_2 = 4$

$y'(0) = c_1 e^0 + c_2(-2)e^0 = -5$

$c_1 - 2c_2 = -5$

$c_1 = 1$

$c_2 = 3$

$y = e^x + 3e^{-2x}$

$y'' - 4y' + 4y = 0, y(0) = 3, y'(0) = 1$ Solution: $\lambda_1 = \lambda_2 = 2$

$\lambda^2 - 4\lambda + 4 = 0$

$(\lambda - 2)^2 = 0$

$\lambda_1 = 2, \lambda_2 = 2$

$y = \underset{3}{\underline{c_1}} e^{2x} + \underset{-5}{\underline{c_2}} x e^{2x}$

$y = (3 - 5x)e^{2x}$

$y(0) = c_1 + 0 = 3 \Rightarrow c_1 = 3$

$y'(0) = 2c_1 e^{2x} + c_2 e^{2x} + c_2 x 2 e^{2x}\Big|_{x=0} = 2 \cdot 3 + c_2 = 1$

$6 + c_2 = 1$

$c_2 = -5$

$y'' - 2y' + 10y = 0$ Solution: $\lambda_1 = 1 + 3i, \lambda_2 = 1 - 3i$

$\lambda^2 - 2\lambda + 10 = 0 \xRightarrow{\text{solve}} \lambda_1 = 1 + 3i$

$\lambda_2 = 1 - 3i$

$\alpha = 1 \quad \beta = 3$

$y = e^x(c_1 \cos 3x + c_2 \sin 3x)$

$y = C e^{\overset{1}{\alpha} x} \cos(\overset{3}{\beta} x) + D e^{\overset{1}{\alpha} x} \sin(\overset{3}{\beta} x)$

Example of Reaction in Batch Reactor

System of linear ODE

$$\begin{cases} \dfrac{d\widetilde{C}_R}{d\tilde{t}} &= -\widetilde{C}_R + K_1\widetilde{C}_I \\ \dfrac{d\widetilde{C}_I}{d\tilde{t}} &= \widetilde{C}_R - K_1\widetilde{C}_I - K_2\widetilde{C}_I \\ \dfrac{d\widetilde{C}_P}{d\tilde{t}} &= K_2\widetilde{C}_I \end{cases} \text{ with BCs } \begin{cases} \widetilde{C}_R(0) &= 1 \\ \widetilde{C}_I(0) &= 0 \\ \widetilde{C}_P(0) &= 0 \end{cases}$$

$R \rightleftarrows I \rightarrow P$

$$\frac{d\tilde{C}_R}{d\tilde{t}} = \frac{d^2\tilde{C}_I}{d\tilde{t}} + k_1\frac{d\tilde{C}_I}{d\tilde{t}} + k_2\frac{d\tilde{C}_I}{d\tilde{t}}$$

start with I

$$\frac{d\widetilde{C}_I}{d\tilde{t}} = \widetilde{C}_R - K_1\widetilde{C}_I - K_2\widetilde{C}_I \implies \widetilde{C}_R = \frac{d\widetilde{C}_I}{d\tilde{t}} + K_1\widetilde{C}_I + K_2\widetilde{C}_I \quad (*)$$

2nd ODE

Substitute $\widetilde{C}_R$ into 1st ODE : $\dfrac{d^2\widetilde{C}_I}{d\tilde{t}^2} + (1 + K_1 + K_2)\dfrac{d\widetilde{C}_I}{d\tilde{t}} + K_2\widetilde{C}_I = 0$

$$\lambda^2 + (1 + K_1 + K_2)\lambda + K_2 = 0$$

$$\lambda_{1,2} = \frac{-(1+K_1+K_2) \pm \sqrt{(1+K_1+K_2)^2 - 4K_2}}{2}$$

$$\widetilde{C}_I = c_1 e^{\lambda_1 \tilde{t}} + c_2 e^{\lambda_2 \tilde{t}} \implies \frac{d\tilde{C}_I}{d\tilde{t}} = c_1\lambda_1 e^{0} + c_2\lambda_2 e^{0}$$

$1 = 1$

$$\widetilde{C}_I(0) = c_1 + c_2 = 0$$

$$\boxed{\widetilde{C}_R(0)} = \left.\frac{d\widetilde{C}_I}{d\tilde{t}}\right|_{\tilde{t}=0} = c_1\lambda_1 + c_2\lambda_2 = 1$$

$$\left.\frac{d\tilde{C}_I}{d\tilde{t}}\right|_{\tilde{t}=0} + k_1\cancel{\tilde{C}_I(0)} + k_2\cancel{\tilde{C}_I(0)}$$

solve.

$$c_1 = \frac{1}{\lambda_1 - \lambda_2} \;\&\; c_2 = \frac{1}{\lambda_2 - \lambda_1}$$

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Continue

$$\tilde{C}_I = \frac{e^{\lambda_1 \tilde{t}} - e^{\lambda_2 \tilde{t}}}{\lambda_1 - \lambda_2}$$

$\frac{d\tilde{C}_I}{d\tilde{t}}$

$$\tilde{C}_R = \boxed{\frac{\lambda_1 e^{\lambda_1 \tilde{t}} - \lambda_2 e^{\lambda_2 \tilde{t}}}{\lambda_1 - \lambda_2}} + (K_1 + K_2) \boxed{\frac{e^{\lambda_1 \tilde{t}} - e^{\lambda_2 \tilde{t}}}{\lambda_1 - \lambda_2}}$$

$\tilde{C}_I$

$$\int \frac{d\tilde{C}_P}{d\tilde{t}} = K_2 \tilde{C}_I = \frac{K_2 \left(e^{\lambda_1 \tilde{t}} - e^{\lambda_2 \tilde{t}} \right)}{\lambda_1 - \lambda_2}$$

$$\tilde{C}_P = \frac{K_2}{\lambda_1 - \lambda_2} \int \left(e^{\lambda_1 \tilde{t}} - e^{\lambda_2 \tilde{t}} \right) d\tilde{t}$$

$$= \frac{K_2}{\lambda_1 - \lambda_2} \left(\frac{e^{\lambda_1 \tilde{t}}}{\lambda_1} - \frac{e^{\lambda_2 \tilde{t}}}{\lambda_2} \right) + c_3$$

$\tilde{t} = 0, \; \tilde{C}_P(0) = 0$

$$\tilde{C}_P(0) = \frac{K_2}{\lambda_1 - \lambda_2} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) + c_3 = 0$$

$$\Longrightarrow \quad c_3 = \frac{K_2}{\lambda_1 \lambda_2}$$

$$\tilde{C}_P = \frac{K_2}{\lambda_1 - \lambda_2} \left(\frac{e^{\lambda_1 \tilde{t}}}{\lambda_1} - \frac{e^{\lambda_2 \tilde{t}}}{\lambda_2} \right) + \frac{K_2}{\lambda_1 \lambda_2}$$

Nth-order Linear Homogeneous ODE with Constant Coefficients

$$\text{General form}: y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1}y' + a_n y = \boxed{0}$$

= Homo case.

$$\text{if } y = e^{\lambda x} \text{ is a solution} \iff \lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n = 0$$

characteristic eqns

$$\lambda : \text{characteristic valus or eigenvalues}$$

solve

$$\lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n = (\lambda - \lambda_1)^{m_1}(\lambda - \lambda_2)^{m_2}\cdots(\lambda - \lambda_s)^{m_s}$$

$$\lambda_1, \cdots, \lambda_s : \text{distinct eigenvalues}$$

$$m_1, \cdots, m_s : \text{multiplicity of respective eigenvaluess}$$

Theorem: Let $\lambda_1, \cdots, \lambda_s$ be the distinct eigenvalues for nth-order linear homogeneous ODE, with multiplicity $m_1, \cdots, m_s$ respectively. Then it has a fundamental set of solutions

if $m_1 = 3$, e^{2x}, xe^{2x}, x^2e^{2x}

$x^2 e^{\lambda_1 x}$ ..

for $\lambda_1 \longrightarrow e^{\lambda_1 x}, xe^{\lambda_1 x}, \cdots, x^{m_1-1}e^{\lambda_1 x};$

for $\lambda_2 \longrightarrow e^{\lambda_2 x}, xe^{\lambda_2 x} \cdots x^{m_2-1}e^{\lambda_2 x}$ ....;

for $\lambda_s \longrightarrow e^{\lambda_s x}, xe^{\lambda_s x}, \cdots, x^{m_s-1}e^{\lambda_s x}.$

$(\lambda - 2)^3(\lambda - 3)^4$

$\lambda_1 = 2$, $m_1 = 3$

$\lambda_2 = 3$, $m_2 = 4$

The general soluton for the homogeneous ODE is the linear combination of fundamental set of solutions: $y_h(x) = \sum_{i=1}^{s}\sum_{j=1}^{m_i} c_{ij}x^{j-1}e^{\lambda_i x}$

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Nth-order Linear Non-homogeneous ODE

non-Homo

$$y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_{n-1}(x)y' + a_n(x)y = f(x)$$

The general solution is in the form: $y(x) = y_h(x) + y_p(x)$

homogeneous solution $y_h(x)$: $y_h^{(n)} + a_1(x)y_h^{(n-1)} + \cdots + a_{n-1}(x)y_h' + a_n(x)y_h = 0$

particular solution $y_p(x)$: $y_p^{(n)} + a_1(x)y_p^{(n-1)} + \cdots + a_{n-1}(x)y_p' + a_n(x)y_p = f(x)$

How to find the particular solution?

1. Method of variation of parameters
2. Method of undetermined coefficients

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Method of Variation of Parameters

$u_1'y_1' + u_1y_1'' + u_2'y_2' + u_2y_2''$

$+ a(u_1y_1' + u_2y_2')$

$+ b(u_1y_1 + u_2y_2) = f$

$(*) \; y_p'' = \frac{d}{dx}(u_1y_1' + u_2y_2')$

$= u_1'y_1' + u_1y_1'' + u_2'y_2' + u_2y_2''$

$$y'' + ay' + by = f(x)$$

set 0

Let y_1 & y_2 be two linearly independent homogeneous solutions

Look for $y_p = \overset{u_1(x)}{u_1} y_1 + \overset{u_2(x)}{u_2} y_2 \Rightarrow y_p' = u_1'y_1 + u_1y_1' + u_2'y_2 + u_2y_2'$

Suppose $u_1'y_1 + u_2'y_2 = 0$ $\quad = y_p' = u_1y_1' + u_2y_2' \quad (*)$

$y_p'' + a y_p' + b y_p = f$

Put y_p into $y'' + ay' + by = f \Longrightarrow \boxed{u_1'y_1' + u_2'y_2' = f}$

$$\begin{cases} u_1'y_1 + u_2'y_2 = 0 \\ u_1'y_1' + u_2'y_2' = f \end{cases} \Longrightarrow u_1' = -\frac{y_2 f}{y_1y_2' - y_1'y_2}, \; u_2' = \frac{y_1 f}{y_1y_2' - y_1'y_2}$$

$$u_1 = -\int \frac{y_2 f}{y_1y_2' - y_1'y_2}\,dx$$

$$u_2 = \int \frac{y_1 f}{y_1y_2' - y_1'y_2}\,dx$$

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Restricted

Example

y_1 is soln to $y''+y=0$
y_2 is " to $y''+y=0$

$$y'' + y = \sec x \qquad f$$

cos x   sin x

$$\boxed{y_1 = \cos x, \; y_2 = \sin x}$$ Homo Solutn.

$$y_1 y_2' - y_1' y_2 = \cos x \cos x - (-\sin x) \sin x = 1$$

$$u_1 = -\int \underbrace{\sin x}_{y_2} \underbrace{\sec x}_{f} \, dx = \ln|\cos x| + c_1$$

$$u_2 = \int \underbrace{\cos x}_{y_1} \underbrace{\sec x}_{f} \, dx = x + c_2$$

$y_p = u_1 Y_1 + u_2 Y_2$

$$y_p = \underbrace{\cos x}_{y_1} \underbrace{\ln|\cos x|}_{u_1} + \underbrace{x}_{u_2} \underbrace{\sin x}_{y_2}$$

$$y = \underbrace{c_1 \cos x + c_2 \sin x}_{\text{Homo}} + \underbrace{\cos x \ln|\cos x| + x \sin x}_{\text{parti}}$$

$$u_1' Y_1 + u_2' Y_2 = \frac{-\sin x}{\cos x} \cdot \cos x + 1 \cdot \sin x = 0.$$

Method of Undetermined Coefficients: Case 1

$$y'' + by' + cy = f(x)$$

$$f(x) = P_n(x)e^{\alpha x}$$

$P_n(x)$: polynomial of degree $n \geq 0$

Look for $y_p = Q(x)e^{\alpha x}$

$$Q'' + (2\alpha + b)Q' + (\alpha^2 + b\alpha + c)Q = P_n(x)$$

$$\begin{cases} \alpha^2 + b\alpha + c \neq 0 & \text{Choose } Q(x) = R_n(x), y = R_n(x)e^{\alpha x} \\ \alpha^2 + b\alpha + c = 0, 2\alpha + b \neq 0 & \text{Choose } Q(x) = xR_n(x), y = xR_n(x)e^{\alpha x} \\ \alpha^2 + b\alpha + c = 0, 2\alpha + b = 0 & \text{Choose } Q(x) = x^2R_n(x), y = x^2R_n(x)e^{\alpha x} \end{cases}$$

Examples

$$y'' - y' - 2y = 4x^2$$

Solution: $y = c_1 e^{2x} + c_2 e^{-x} - 3 + 2x - 2x^2$

Examples

$$y''' + 2y'' - y' = 3x^2 - 2x + 1$$

Solution: $y = c_1 + c_2 e^{(-1+\sqrt{2})x} + c_3 e^{(-1-\sqrt{2})x} - 27x - 5x^2 - x^3$

Examples

$$y'' - 2y' + y = xe^x$$

Solution: $y = c_1 e^x + c_2 x e^x + \frac{1}{6} x^3 e^x$

Method of Undetermined Coefficients: Case 2

$y'' + by' + cy = f(x)$ $\quad f(x) = P_n(x)e^{\alpha x}\cos(\beta x)$ or $f(x) = P_n(x)e^{\alpha x}\sin(\beta x)$

Look for y_p s.t. $\quad y'' + by' + cy = P_n(x)e^{(\alpha + i\beta)x}$ [*]

Using method in Case 1 to obtain: $\quad z(x) = u(x) + iv(x)$

Substitute $z(x)$ into [*] $\quad u(x)$ is solution of $y'' + by' + cy = P_n(x)e^{\alpha x}\cos(\beta x)$

$v(x)$ is solution of $y'' + by' + cy = P_n(x)e^{\alpha x}\sin(\beta x)$

Method of Undetermined Coefficients: Case 2

Example $$y'' - 2y' + 2y = e^x \cos x$$

$$y = c_1 e^x \cos(x) + c_2 e^x \sin(x) + \frac{x e^x \sin x}{2}$$

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Method of Undetermined Coefficients: Case 2

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Announcements

- No homework this week
- Homework for week 2, 5, 9, 10, 11 with each 2% for the final grade
- Read "Guidelines for Mid-term Project" in eDimension
- Next week CNY: No class